Exercises: Determinants, Orthogonal Matrices, and Orthonormal Bases

Mathematical Foundations for Machine Learning

Determinant Properties and Volume Interpretation

Exercise 1.1 [Easy]

Let $A, B \in \mathbb{R}^{n\times n}$. Using only the determinant product rule $\det(AB) = \det(A)\det(B)$, prove the following properties:

(a) $\det(A^{-1}) = \frac{1}{\det(A)}$ for invertible A.
(b) $\det(\alpha A) = \alpha^n \det(A)$ for $\alpha \in \mathbb{R}$.
(c) If A is orthogonal, then $|\det(A)| = 1$.

$\det(A^T) = \det(A)$

squared matrixes

Exercise 1.2 [Easy]

Let $R \in \mathbb{R}^{4\times 4}$ be an upper triangular matrix with diagonal entries $r_{11} = 2$, $r_{22} = -3$, $r_{33} = 1$, $r_{44} = 5$.

(a) What is $\det(R)$?
(b) What is $\det(R^T R)$?
(c) Is R invertible? If yes, what is $\det(R^{-1})$?

- det Q = ±1

Exercise 1.3 [Medium]

Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

(a) Compute $\det(A)$. Interpret this geometrically: if $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form the unit square, what is the area of the parallelogram spanned by Av_1 and Av_2?

(b) Let $B = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Without computing AB explicitly, what is $\det(AB)$? What is the volume scaling factor when applying first B then A?

(c) In a 2D generative model, a linear transformation T maps a latent distribution supported on the unit square to the output space. If T has $\det(T) = 0.5$, explain what this means for the density of the output distribution relative to a uniform input density. (Hint: The change-of-variables formula involves $|\det(J)|^{-1}$ for the density.)

density = 2 means that the probability will be more concentrated because we halved the area with det(T) = 0.5

Exercise 1.4 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be invertible and let $u, v \in \mathbb{R}^n$. The **matrix determinant lemma** states:

$$\det(A + uv^T) = (1 + v^T A^{-1} u)\det(A).$$

(a) Using this lemma, compute $\det(I_n + uv^T)$ for arbitrary $u, v \in \mathbb{R}^n$.
(b) Let $v \in \mathbb{R}^n$ with $\|v\|_2 = 1$. Compute $\det(I_n + vv^T)$ and $\det(I_n - vv^T)$.
(c) In ridge regression, the regularized solution involves $(X^T X + \lambda I)^{-1}$. Suppose $X = uv^T$ is a rank-1 data matrix. Use part (a) to express $\det(X^T X + \lambda I)$ in terms of λ, u, and v.

$w^* = (X^T X + \lambda I)^{-1} X^T y$

with lambda > 0 you always have invertible in ridge regression, even with rank one data

Exercise 1.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be invertible. We define the map $T_A : \mathbb{R}^n \to \mathbb{R}^n$ by $T_A(x) = Ax$.

(a) Let $S \subset \mathbb{R}^n$ be a measurable set with volume $\mathrm{vol}(S)$. Show that $\mathrm{vol}(T_A(S)) = |\det(A)| \cdot \mathrm{vol}(S)$. You may use the substitution formula for integrals: $\int_{T_A(S)} 1\, dy = \int_S |\det(A)|\, dx$.

$$\int_{T_A(S)} 1\, dy = \int_S 1 \cdot |\det(A)|\, dx = |\det(A)| \int_S 1\, dx = |\det(A)| \cdot \mathrm{Vol}(S)$$

(b) Suppose $A = QR$ is the QR decomposition where Q is orthogonal and R is upper triangular with positive diagonal entries $r_{11}, \ldots, r_{nn}$. Show that $|\det(A)| = \prod_{i=1}^{n} r_{ii}$.

(c) In normalizing flows, one composes K invertible layers f_k with Jacobians J_k. Using the chain rule for determinants, express $\log|\det(J_{\text{total}})|$ in terms of the individual Jacobians. Why is the triangular Jacobian structure (as in autoregressive flows) computationally attractive?

Orthonormal Bases and Orthogonal Matrices

Exercise 2.1 [Easy]

Let $Q \in \mathbb{R}^{n\times n}$ be an orthogonal matrix.

(a) Prove that Q preserves inner products: $\langle Qu, Qv\rangle = \langle u, v\rangle$ for all $u, v \in \mathbb{R}^n$.

(b) Prove that Q preserves norms: $\|Qu\|_2 = \|u\|_2$ for all $u \in \mathbb{R}^n$.

(c) If λ is an eigenvalue of Q (possibly complex), show that $|\lambda| = 1$.

Exercise 2.2 [Medium]

Consider $\mathbb{R}^3$ with the standard inner product. Let

$$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\-1\\2\end{pmatrix}, \quad v_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}.$$

(a) Verify that $\{v_1, v_2, v_3\}$ is an orthonormal basis of $\mathbb{R}^3$.

c = B1 * v -> change of basis

(b) Express $w = \begin{pmatrix}3\\1\\2\end{pmatrix}$ in the basis $\{v_1, v_2, v_3\}$, i.e., find coefficients $\alpha_1, \alpha_2, \alpha_3$ such that $w = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$.

if you take inner product with v1 in both sides, you get <w, v1> = alfa bec. orth. b.

(c) Let $Q = (v_1 \mid v_2 \mid v_3)$ be the matrix with columns v_1, v_2, v_3. Verify that $Q^T w$ gives the coordinate vector of w in this basis. Why does this work?

Exercise 2.3 [Medium]

Let $Q \in \mathbb{R}^{n\times n}$ be orthogonal and $A \in \mathbb{R}^{n\times n}$ be symmetric.

(a) Show that $Q^T AQ$ is symmetric.

choose w = Q^tv and multiply by it

(b) Show that A and $Q^T AQ$ have the same eigenvalues.

directions rotated but magnitudes preserved

(c) Show that $\|Ax\|_2 = \|Q^T AQy\|_2$ when $x = Qy$. Interpret: does the orthogonal change of basis affect the "behavior" of the linear map A?

Q^tAQ is like rotating the matrix A, applying the transofrmation y (rotated x) and then rotating back,so the norm will be equal

Exercise 2.4 [Hard]

Let $P \in \mathbb{R}^{n\times n}$ be a matrix satisfying $P^2 = P$ and $P^T = P$ (i.e., P is an orthogonal projection).

(a) Show that the only possible eigenvalues of P are 0 and 1.

(b) Let $r = \operatorname{rank}(P)$. Show that $\operatorname{tr}(P) = r$.

(c) Show that $I - P$ is also an orthogonal projection. What subspace does it project onto?

(d) In the context of PCA, suppose we project data onto the top-r principal components. The projection matrix is $P = U_r U_r^T$ where $U_r \in \mathbb{R}^{n\times r}$ has orthonormal columns. Verify that $P^2 = P$ and $P^T = P$, and show that $\operatorname{tr}(P) = r$.

Gram-Schmidt, Orthogonal Projection, QR Decomposition, and Change of Basis

Exercise 3.1 [Medium]

Apply the Gram-Schmidt process to the vectors

$$u_1 = \begin{pmatrix}1\\1\\0\end{pmatrix}, \quad u_2 = \begin{pmatrix}1\\0\\1\end{pmatrix}, \quad u_3 = \begin{pmatrix}0\\1\\1\end{pmatrix}$$

to obtain an orthonormal basis $\{v_1, v_2, v_3\}$ of $\mathbb{R}^3$.

normalize after the substraction / projection

(a) Compute v_1.
(b) Compute v_2 by removing the component of u_2 along v_1 and normalizing.
(c) Compute v_3 by removing the components of u_3 along v_1 and v_2, then normalizing.
(d) Verify that the matrix $Q = (v_1 \mid v_2 \mid v_3)$ is orthogonal.

Exercise 3.2 [Medium]

proj(b into a) = <a, b> / <a, a> -> the path from b into ac will be orthogonal to a when closest. Then that generalizes to the matrix formula

Let $W = \text{span}\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}\right\} \subset \mathbb{R}^3$.

(a) Find the orthogonal projection matrix P_W onto W. (Hint: If U has columns forming a basis of W, then $P_W = U(U^TU)^{-1}U^T$.) if you want to project any vector b, add the b to the formula

(b) Compute the orthogonal projection of $b = \begin{pmatrix}3\\3\\0\end{pmatrix}$ onto W.

(c) Find the distance from b to the subspace W, i.e., $\|b - P_W b\|_2$.

(d) In least-squares regression, $\hat{y} = X(X^TX)^{-1}X^Ty$ is the projection of y onto $\text{col}(X)$. The residual $y - \hat{y}$ is orthogonal to $\text{col}(X)$. Verify this orthogonality property for your computation above: check that $b - P_W b \perp w$ for each basis vector w of W.

Exercise 3.3 [Medium]

when solving directly linear regression, we are computing the orthogonal projecitonof the s

Find the QR decomposition of the matrix

$$A = \begin{pmatrix}1 & 1 & 0\\1 & 0 & 1\\0 & 1 & 1\end{pmatrix}.$$

(a) Apply Gram-Schmidt to the columns of A to find the orthogonal matrix Q.
(b) Compute $R = Q^TA$ and verify it is upper triangular with positive diagonal.
(c) Verify that $|\det(A)| = \prod_i r_{ii}$.
(d) Use the QR decomposition to solve $Ax = b$ for $b = \begin{pmatrix}2\\1\\3\end{pmatrix}$ via back-substitution on $Rx = Q^Tb$.

Ax = b -> QRx = b -> Rx = Q^tb

Exercise 3.4 [Medium]

Let $B_1 = \{e_1, e_2\}$ be the standard basis of $\mathbb{R}^2$ and let $B_2 = \left\{w_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, w_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix}\right\}$.

(a) Verify that B_2 is an orthonormal basis.
(b) Find the transition matrix Q from B_1 to B_2 such that $[x]_{B_2} = Q[x]_{B_1}$. (Hint: $Q = W^TV$ where V, W have the respective basis vectors as columns.)
(c) Verify that Q is orthogonal and compute $\det(Q)$. What geometric transformation does Q represent?
(d) Given the vector $x = \begin{pmatrix}3\\1\end{pmatrix}$ in B_1, find its coordinates in B_2.

with matrix and column by column approach:- B1 -Q-> B2 then if v in B1 : v = Qv_b2 -> e1 = 1w

Mixed Challenging Exercises

Exercise 4.1 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigendecomposition $A = Q\Lambda Q^T$, where Q is orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

(a) Show that $\det(A) = \prod_{i=1}^n \lambda_i$ and $\mathrm{tr}(A) = \sum_{i=1}^n \lambda_i$.

(b) For the matrix $A = 2vv^T + 3I$ where $v \in \mathbb{R}^n$ with $\|v\| = 1$, find all eigenvalues, $\det(A)$, $\mathrm{tr}(A)$, and all singular values.

(c) Compute $\max_{\|x\|=1} x^T Ax$ and $\max_{\|x\|=1} \|Ax\|_2$ for the matrix in part (b).

(d) Suppose we apply gradient descent on the quadratic $f(x) = \frac{1}{2}x^T Ax$ with A from part (b). The condition number is $\kappa = \lambda_{\max}/\lambda_{\min}$. Compute κ and discuss how the convergence rate depends on it.

Exercise 4.2 [Hard]

Let $u_1, \ldots, u_n$ be an orthonormal basis of $\mathbb{R}^n$ and define $P_k = u_k u_k^T$ for each k.

(a) Show that $\sum_{k=1}^n P_k = I_n$ (the completeness relation / resolution of the identity).

(b) Show that $P_j P_k = 0$ for $j \neq k$ and $P_k^2 = P_k$.

(c) For a symmetric matrix $A = \sum_{k=1}^n \lambda_k P_k$ (spectral decomposition), derive $\det(A)$ and $\mathrm{tr}(A)$ using the properties from (a) and (b).

(d) Using the resolution of the identity, show that for any $x \in \mathbb{R}^n$:

$$\|x\|^2 = \sum_{k=1}^n |\langle x, u_k\rangle|^2$$

(Parseval's identity). Explain why this is relevant for signal compression in ML.

Exercise 4.3 [Hard]

Let $X \in \mathbb{R}^{n\times d}$ be a data matrix with n samples and d features. The thin QR decomposition is $X = QR$ where $Q \in \mathbb{R}^{n\times d}$ has orthonormal columns and $R \in \mathbb{R}^{d\times d}$ is upper triangular.

(a) Show that the least-squares solution $\hat{\theta} = (X^T X)^{-1} X^T y$ can be rewritten as $\hat{\theta} = R^{-1} Q^T y$.

(b) Show that the projection matrix onto $\mathrm{col}(X)$ satisfies $P = QQ^T$ (i.e., it depends only on Q, not on R).

(c) Show that $\det(X^T X) = (\det R)^2 = \left(\prod_{i=1}^d r_{ii}\right)^2$.

(d) In numerical linear algebra, solving normal equations via $(X^T X)^{-1} X^T y$ can be numerically unstable because $\kappa(X^T X) = \kappa(X)^2$. Explain why the QR-based approach $\hat{\theta} = R^{-1} Q^T y$ is more stable.